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Chapter 5

Forensic Science as a Social Process

Introduction

Forensic science has become an integral part of the criminal justice system. Yet it is often discussed in relative isolation. This chapter adopts a holistic and systemic perspective to explore the role of forensic science in the criminal justice system. It views criminal investigations as social processes and examines how forensic science is enacted at various stages in the criminal justice process from crime scene to court, emphasising the social construction of forensic evidence. The chapter examines the decisions that are made about the use of forensic science at critical phases in a criminal investigation. These include decisions at crime scenes, in forensic fieldwork, at forensic laboratories, in police investigations, during court preparation, and in the courts. The chapter argues that from a critical forensic studies perspective, all actors in the criminal justice system need to understand their roles in the context of the wider criminal justice system and develop the ‘forensic literacy’ necessary for their roles. This broader contextual knowledge can contribute to asking the ‘right’ questions, identifying the relevant expertise, and drawing upon this expertise in a manner that enhances the likelihood that criminal investigations will contribute to ‘just’ outcomes.

The Social Construction of Forensic Evidence

Opposing narratives about forensic science have been prominent (McCartney, forthcoming). On the one hand, forensic science has enabled a new science-led approach to policing and has ‘revolutionised’ the criminal justice system. On the other hand, reviews of miscarriages of justice have identified forensic science as a significant cause of wrongful convictions (Dioso-Villa, 2015b; Garrett, 2011; Garrett & Neufeld, 2009; Gould, Carrano, Leo, & Young, 2012; Smit, Morgan, & Lagnado, 2018) and a number of prominent reports have been highly critical of the current status of forensic science (National Research Council (NRC), 2009; President’s Council of Advisors on Science and Technology, 2016). Both narratives are partially ‘true’. Forensic science is an *applied* science; it cannot meet the expectations of an idealised version of ‘pure’ science, nor the unrealistic public expectations generated by crime shows and media representation. In reality, forensic science is always enacted by people in specific social contexts (Williams & Weetman, 2012).

It was not until the turn of the twenty-first century that interest in the social processes through which forensic science is enacted began to emerge. In the United Kingdom, the Home Office published a ground-breaking report in 2000 that focused on the use of forensic sciences within police services (Her Majesty’s Inspectorate of Constabulary, 2000). In the

same year in Australia, a study was commissioned by the National Institute of Forensic Science to examine the level of knowledge and awareness of forensic science among general duties police officers in three jurisdictions (Julian, 2005). In the United States, the National Institute of Justice (NIJ) identified a gap in knowledge ‘regarding how human beings can and should use new forensic technologies’, and in 2005 established a new portfolio on ‘Social Science Research on Forensic Science’ to examine ‘how the explosion of new forensic technologies and techniques was affecting the criminal justice system’ (Browning, 2015, p. 41). By this time, socio-legal scholars had already begun to critique legislative provisions and courtroom practices surrounding expert evidence and the admissibility of various forensic science ‘disciplines’ (Roberts, 1994; Edmond & Mercer, 1997). This work collectively lay the foundations for the emergence of a more critical social science research agenda on the role of forensic science (Lynch, 1998; Cole, 2010a; Thompson, 2008); however, early critiques rarely adopted a systemic perspective. For critical forensic studies, examining these underlying social processes is critical to understanding how forensic science works in the criminal justice system.

One of the key insights from scholars in social science is that forensic ‘evidence’ is a social construction rather than an objective ‘fact’. Drawing on Appadurai’s (1986) *The Social Life of Things*, Corinna Kruse (2016) has produced a comprehensive discussion of this process in *The Social Life of Forensic Evidence*. As she argues: ‘forensic evidence is produced by the judicial system as well as by wider society, through sociocultural practices as much as through scientific ones, and ... these practices merit attention’ (Kruse, 2012, p. 308).

The first step in understanding forensic evidence as a social construction is to acknowledge that the materials collected at crime scenes are ‘traces’ of a crime rather than evidence *per se*. Traces are therefore remnants of activities and people. A vast amount of trace material exists at a crime scene; it only becomes meaningful as forensic *evidence* if it has been collected, processed and interpreted as relevant to the criminal investigation. In this way, traces are ‘transformed into legally meaningful evidence’ (Kruse, 2012, p. 307; see also Margot, 2011). This view contrasts with the common (mis)representation – perpetuated by television crime shows – of forensic evidence as an objective ‘fact’ just waiting to be discovered at the crime scene. On the contrary, forensic evidence is socially constructed; traces are transformed into forensic evidence by rendering them meaningful to the case in question. Thus, the forensic process is a *meaning-making process*.

The social construction of forensic evidence is a complex process that involves interpretations made by numerous actors in various agencies as the trace travels through the criminal justice system. As noted by Kruse, the trajectory through which forensic evidence and criminal investigations are socially constructed ‘can be said to span the links of the legal chain, and it involves plaintiffs, suspects, witnesses, crime scene technicians, forensic scientists, police investigators, prosecutors, defence lawyer, judges, and lay assessors’ (Kruse, 2016, cited in Costa & Santos, 2019, p. 471). Dahl (2007) uses the metaphor of a jigsaw puzzle to describe the process through which evidence (in this case DNA) is socially

constructed by numerous actors – whom he calls ‘construction workers’ – from offenders through to judges and the jury:

All of these ‘construction workers’ may influence how the piece of jigsaw puzzle consisting of DNA evidence is shaped, presented and perceived in a trial. Despite this, DNA is often presented and perceived as an objective truth; a piece of jigsaw puzzle that has only one given size, shape and form. (Dahl, 2007, as cited in Santos, 2014, p. 183)

The meaning and value of traces such as DNA varies across cases, over time and in different situational contexts. By the time a trace reaches court as evidence, it will have been interpreted and re-interpreted many times until it has been imbued with specific meanings of relevance to the process of adjudication. From this perspective, the meaning of forensic evidence is not a ‘given’; it is contestable and can change over time. Forensic objects should therefore be viewed as always in a state of ‘evidence to be’ until the court proceedings have been concluded (Costa & Santos, 2019).

The Criminal Justice System

Forensic science is also a set of practices embedded in an institutional framework; namely, the criminal justice system (Fraser & Williams, 2009b). The socio-cultural practices described above take place within a variety of institutional settings, with different professional communities; each with their own culture of knowledge (epistemology) or, what Knorr-Cetina has called ‘epistemic cultures’. She defines these as:

amalgams of arrangements and mechanisms – bonded through affinity, necessity, and historical coincidence – which, in a given field, make up how we know what we know. Epistemic cultures are cultures that create and warrant knowledge. (Knorr-Cetina, 1999, as cited in Costa & Santos, 2019, p. 471)

Table 5.1 depicts the epistemic cultures and institutional locations of practitioners typically involved in different phases of the criminal justice process. While traditionally each community of practitioners may maintain a dominant epistemic culture, research has identified new epistemic cultures emerging among some practitioners (e.g., an intelligence culture among crime scene examiners (CSEs) and police investigators), and the boundaries between epistemic cultures may be more or less porous in some contexts (e.g., more so between the crime scene and the police investigation than between the forensic laboratory and the police investigation). Challenges may arise when multiple epistemic cultures co-exist in the same institutional location and/or phase.

Table 5.1 Epistemic cultures at various phases in the criminal justice process

Criminal justice phase	Institutional location	Roles	Epistemic culture(s)^
Crime scene examination	• Police organisation • Police forensic services/forensic laboratory	• First responders (e.g., police officers) • Forensic field scientists (e.g., CSEs) • Forensic specialists (e.g., fire investigators)	• Hunch culture* (intuition-based) • Intelligence culture (pattern-oriented) • Scientific culture (research-informed)

Forensic analysis	• Forensic laboratory • Police forensic services	• Forensic laboratory scientists (e.g., DNA analysts) • Forensic field scientists (e.g., fingerprint experts)	• Bubble culture* (boundary-driven) • Scientific culture (research-informed) • Police culture (conviction-centred)
Police investigation	• Police organisation	• Police investigators and detectives	• Hunch culture* (intuition-based) • Police culture (conviction-centred) • Intelligence culture (pattern-oriented) • Scientific culture (research-informed)
Pre-trial and trial processes	• Legal system (i.e., Office of the Public Prosecutor; defence offices) • Court	• Public prosecutors • Defence lawyers • Judges • Forensic scientists • Jurors	• Legal culture (tradition-centred) • Office culture* (legal-translation – inquisitorial system) • Game culture** (legal-advantage – adversarial system) • Scientific culture (research-informed)

Sources:

*Costa & Santos (2019)

**Chapman (2015)

^Notes:

A brief description of the epistemic cultures listed in Table 5.1:

- *hunch culture* – reliant on intuition-based reasoning or sensemaking; draws on past knowledge and experience (e.g., knowledge of local offenders);
- *intelligence culture* – oriented towards the identification of a pattern or series (e.g., searching for links across a number of incidents);
- *scientific culture* – informed by research and validation in pursuit of accurate, reproducible and reliable knowledge; knowledge may change over time (e.g., empirical research used to assess the probability of alternative hypotheses);
- *bubble culture* – driven by a desire to maintain boundaries between communities of practitioners and different epistemic cultures (e.g., to reinforce status and claims of specialised knowledge);
- *police culture* – centred on crime detection and crime reduction through convictions (e.g., investigations are case-based and focused on identification of the offender);
- *legal culture* – driven by precedent and tradition and informed by legal jurisprudence (e.g., searching for certainty and ‘truth’);
- *office culture* – focused on translating evidence into legal language for the benefit of legal actors (e.g., forensic traces and/or samples become forensic ‘evidence’ through the language of the law);
- *game culture* – focused on gaining legal advantage (e.g., driven by the goal of ‘winning’ against one’s adversary in court).

As the trajectory of the trace travels through various institutions with their own epistemic cultures, ‘[t]he same crime can generate different stories, different narratives about what happened, according to the actors and epistemic cultures who intervene’ (Costa & Santos, 2019, pp. 471–472). The construction of these narratives involves actors in different institutional settings assessing, valuing and interpreting the forensic evidence in different ways. This includes differential assessments of the value of the evidence, as actors draw on

different epistemic cultures ‘set in a framework of divergent institutional goals’ (Santos, 2014, p. 189; see also Brookman, Jones, Williams, & Fraser, 2020).

It is through these different meanings and interpretations – and the actions that are based on them – that ‘the boundary lines of distinct epistemic cultures’ are continuously redrawn (Costa & Santos, 2019, p. 472). Actors in different institutional settings are continuously engaged in ‘boundary work’, whereby they establish and reinforce their expertise through a focus on the ‘differentiating features of their knowledge’ (Costa & Santos, 2019, p. 476). This boundary work impacts on the ways in which forensic traces and/or evidence are interpreted, evaluated and utilised in everyday practices. For example, in a study of forensic DNA phenotyping (FDP), Hopman (2020) shows how the technology is perceived and applied differently in research laboratories, forensic laboratories and criminal investigations, with each setting having its own logic. Problems that arise from – or within – alternative epistemic cultures may also be highlighted in this process; for example, identifying a failure among forensic scientists to adhere to adequate forensic procedures or the impact of contextual bias on the ‘impartial’ results of forensic analyses; the concealment of forensic evidence by police investigators; or the damaging interference of personal relationships and the organisational biases that might influence the course of an investigation (see, e.g., the analysis of the Saltão case in Portugal by Costa & Santos, 2019).

At various points along the trajectory from crime scene to court, these often divergent interpretations may lead to contestation; most notably, for example, in the courts in an adversarial system. However, contestation may also occur at other points when forensic information requires translation across epistemic cultures; for example, at the crime scene, when CSEs and police are making decisions about the most appropriate traces to collect, or during the triage process at the laboratory, when decisions are being made about which traces to analyse and the order of priority for these analyses. While these are often described as ‘problems of communication’ they may point to a more underlying systemic issue associated with the boundary work involved in maintaining expertise within distinct epistemic cultures. Kruse (2020) refers to these as ‘epistemic frictions’.

Importantly, however, what happens to the forensic evidence in one institutional setting has implications for its trajectory through the other institutional settings. As Morgan has noted:

The actions and decisions taken at the crime scene will have an impact on the questions asked of the evidence and therefore the type of analysis undertaken on exhibits at the laboratory, the approach taken to make inferences about the results of the analysis, and how these conclusions are presented as intelligence and/or evidence. (Morgan, 2017a, p. 457)

Along this trajectory from crime scene to court, the narratives accumulate and are translated across epistemic cultures so that ‘[e]ach piece of forensic evidence is a combination of the different epistemologies of which it has been a part during its social life’ (Kruse, 2016, as cited in Costa & Santos, 2019, p. 472). These narratives ‘are told and re-told ... as they are assembled, arranged, re-arranged, and shaped into one coherent, plausible narrative’ presented in the court (Brookman et al., 2020, p. 20). In this way, forensic

‘evidence’ is co-constructed by actors in the different epistemic cultures in the criminal justice system (Costa & Santos, 2019). This highlights the importance of adopting a holistic, systemic approach to the analysis of forensic evidence that examines the whole process from crime scene to court (Morgan, 2017a, 2017b).

The Forensic Process in the Criminal Justice System: A Systemic Framework

One way to examine the role of forensic evidence in the criminal justice system is to follow the evidence through the various stages of the criminal justice process from crime scene to court and to critically examine the underlying social processes, including:

- the detection and collection of traces at the crime scene (such as paint flecks, hair, fibres from clothing, weapon marks);
- the analysis of these traces at the laboratory;
- the extent to which traces and results of forensic analyses influence police investigations (are they used to exonerate suspects early or, more typically, do they simply add weight to the prosecution’s case?);
- how forensic science information is used by lawyers (is it as valuable to the defence as it is to the prosecution?);
- how forensic evidence is presented in court (what is considered admissible? Who presents it? What criteria are applied to determine the scientific and probative value of the evidence presented?);
- how juries understand forensic evidence and in what ways this influences their deliberations; and
- whether there is an overreliance on forensic evidence in the courts and, most importantly, whether the reliance on forensic evidence improves the likelihood of justice for the accused (Julian & Kelty, 2012, p. 1).

This can be termed ‘the forensic process’; that is, the trajectory of forensic material from the collection of traces at the crime scene through to their interpretation as forensic evidence and subsequent presentation in court (Julian & Kelty, 2015). The forensic process can be conceptualised as a series of phases through the criminal justice system. It is in the problems that can occur in these social processes that we see the origins of unjust outcomes such as miscarriages of justice – including, but not limited to, wrongful convictions. Figure 5.1 depicts the forensic process from the crime scene to the court within a systemic framework. This is necessarily an indicative model that glosses over the complexities inherent in the forensic process as it is enacted in specific institutional contexts (e.g., different jurisdictions).

INSERT FIGURE 5.1 HERE – ‘The Forensic Process in the Criminal Justice System’

The use of a systemic approach that focuses on the social construction of forensic evidence – including the decisions made by various actors – enables the identification of critical points in the forensic process that may lead to errors that contribute to miscarriages of justice. The criminal justice system has been criticised in the past for not being open to a ‘learning from error’ approach (Doyle, 2010). Rather, there has been a tendency to take a ‘bad apples’ approach, which views errors as being caused by individual decision makers, and aims to minimise knowledge and awareness of these errors by others in the system. By contrast, within a systemic framework, miscarriages of justice are viewed ‘not as anomalies in the criminal justice system, but as the result of systemic errors that can occur at various stages and by different actors in the system’ (Dioso-Villa, Julian, Kebbell, Weathered, & Westera, 2016, p. 158).

The systemic perspective that follows was developed in an Australian research project that explored processes and outcomes from the crime scene to the court. It examined how effectively various actors within the criminal justice system used forensic information and forensic services (Julian, et al., 2011). The project adopted a forensic-led case-processing framework that explored the decisions made about the use of forensic evidence during the successive phases from crime scene to court. In particular, the project highlighted the significance of initial decisions made at the crime scene for case progression (Kelty, Julian, & Hayes, 2015).

The Crime Scene

Forensic science begins at the crime scene. Effective processing of a crime scene is one of the most critical aspects of criminal investigations (Julian, Kelty, & Robertson, 2012; Wyatt, 2014). While this may appear to be an obvious statement, forensic science has not always been conceptualised as encompassing crime scene work (Crispino, 2008; Julian et al., 2012; Laurin, 2013; Cole, 2013). Traditionally, the activities undertaken at the crime scene were viewed as technical – namely, bagging and tagging – rather than scientific (Wyatt, 2014). The role of CSE was viewed as police work in contrast to the work of forensic scientists, which was understood to occur at the laboratory. However, the crime scene – and more specifically the role of the CSE – has come under increased scrutiny since the turn of the twenty-first century leading to a growing interest in, and indeed concern with, the ways in which crime scenes are examined and traces are identified, collected and stored.

The role of the CSE is to search for traces of a potentially criminal activity. A trace can be defined as:

a pattern, a signal or material transferred during an event (often unknowingly by the actors of the event). It is the remnant (the memory) of the source (identity – who, with what?) and of the activity (what, how, when, why?) that produced it. (Margot, 2011, p. 91)

The term ‘trace’ denotes many different kinds of material including, but not limited to, glass fragments, fibres, bloodstains, shoeprints, fingerprints and DNA samples. The first step in the

forensic process is thus to decide which traces are *forensic* traces; in other words, which traces are relevant (or potentially relevant) to the case. Furthermore, forensic traces are meaningful in various ways. As Margot (2011, p. 91) explains: ‘the type of information that is brought by traces is not only its potential for identification, but also for reconstruction and association (trace-to-person, person-to-trace, trace-to-trace and person-to-person)’.

Crime scene examination involves decision making based on interpretations of traces at crime scenes. As de Gruijter, Nee, & de Poot show, this broader conception of crime scene work requires much more than technical skills:

CSIs have to perceive the visible or latent traces that are left at the crime scene, and have to infer their causes. Traces alone do not provide enough information to infer a whole series of events that has taken place; they only provide clues to their inferences. Interpretations are necessary in order to give meaning to these clues, to link them to actions and to construct a story about the crime. (2017, p. 421)

Several critical issues at the crime scene influence the construction of the emerging narrative. These include questions such as: What traces do CSEs collect? How well do they collect traces, and for what purposes? What makes a good CSE? What knowledge and skills do they need? How should they be trained?

A number of factors influence the interpretations and decisions made by CSEs and thus the narratives that they construct. First, crime scene organisational models vary across jurisdictions. In most jurisdictions, crime scenes are attended by police personnel with the addition of external forensic specialists when this is considered valuable and/or necessary. A distinction is usually made between *volume crime* (e.g., break and enter, motor vehicle theft) and serious or *major crime* (e.g., sexual assault, homicide), with the personnel attending major crimes having greater access to resources than those who attend volume crimes. The status of CSEs also varies across jurisdictions; for example, they can be sworn police officers with either minimal or extensive training in forensic science (the latter often designated as scene of crime officers (SOCOs)) or civilians with a science degree (Roux, Julian, Kelty, & Ribaux, 2014). CSEs in Scotland, England and Wales are increasingly likely to be civilian scientists (Ludwig, Fraser, & Williams, 2012; Wilson-Kovacs, 2014; Wyatt, 2014), whereas in Sweden and most Australian jurisdictions they are likely to be sworn police officers (Kruse, 2020), and in Switzerland they are either civilians or sworn police officers (Resnikoff, Ribaux, Baylon, Jendly, & Rossy, 2015).

Second, the role of CSEs in the criminal investigation varies. Williams (2004) categorises two broad approaches: CSEs providing ‘technical assistance’, who perform clearly defined roles and follow standard procedures, versus CSEs working as ‘expert collaborators’, who are fully integrated into the investigative process as professionals making informed decisions on a case-by-case basis (Tilley & Townsley, 2009). This distinction is variously described as ‘process’ versus ‘expert’ integration (Fraser, 2000) or ‘procedural’ versus ‘discretionary’ models (Burrows, Hopkins, Robinson, Speed, & Tilley, 2005). A number of reviews into the efficiency and effectiveness of CSE work have recommended the adoption of the latter approach; however, numerous studies have found that this vision of

how CSEs *could* work most effectively is rarely realised in practice, especially in volume crime investigations (Bradbury & Feist, 2005; Ludwig et al., 2012; Wilson-Kovacs, 2014).

Third, studies have found a high degree of variation in CSE performance, with some CSEs outperforming their peers in terms of the number of crime scenes they attended, the time taken to collect traces and submit them to the laboratory for analysis, and the quality of the traces they collected, in that they were more likely to lead to positive identifications (Home Office, 2007; Ludwig, Edgar, & Maguire, 2014). In Australia, Kelty, Julian and Robertson (2012) identified that the key skill sets leading top performing CSEs to excel in their work included a wide range of social skills in the following categories: knowledge base, experience, work orientation, approach to life, communication, professional demeanour, and cognitive abilities. Studies such as these have led researchers and practitioners to call for the professionalisation of CSEs through the development of entry qualifications, training, regulation and a research culture (Robertson, White, Kelty, & Julian, 2014; Kelty, 2012; Kelty et al., 2012; Kelty & Gordon, 2012). More recently, there has been an emergence of sociological studies that have adopted case study and ethnographic methods to explore the *actual* work of CSEs and other actors in the forensic process (Wyatt, 2014; Wilson-Kovacs, 2014; Kruse, 2020; Machado & Granja, 2019; Santos, 2014; Costa & Santos, 2019). These process-oriented studies show the complexity of the work of CSEs and, importantly, highlight critical points at which potential errors may occur that can contribute to miscarriages of justice. Case Study 5.1 describes one such study conducted in Australia.

Case Study 5.1: The Effectiveness of Forensic Science in the Criminal Justice System

An Australian project identified and tracked the key decision-making points from the collection of traces at the crime scene through to the courts (Julian et al., 2011). Through an in-depth analysis of 11 case studies of serious crime (homicide and fatal arson) the project identified critical decision-making points in the investigative phase that resulted in either the effective or ineffective use of forensic expertise or forensic evidence. These critical decision-making points include:

- determining whether and when to seal an incident scene and declare it as a crime scene;
- deciding whether to call forensic specialists to assist in the crime scene examination, as well as determining which forensic experts are required;
- ensuring that forensic experts remain within the boundaries of their knowledge/expertise and that the collection of forensic traces is not contaminated through flawed practices;
- ensuring the timely submission of relevant physical evidence and trace material to the laboratory for analysis;
- determining whether to hold a case conference or investigative meeting/briefing to inform effective triage at the laboratory; and

- deciding if and when to take advice from forensic experts regarding the benefits and limitations of conducting various forensic analyses, as this will affect the information upon which police are able to rely as the investigation progresses (Kelty et al., 2015).

The extent to which these critical decisions at the crime scene are based on science is contested. In their analysis of the differing epistemic cultures through which forensic artefacts travel in the course of a criminal investigation in Portugal, Costa and Santos (2019, p. 472) describe police officers as ‘the first elements in the chain of custody’ and ‘responsible for constructing the first narrative of the criminal event’. This narrative is constructed in the context of a *hunch culture*; that is, one that relies on intuition-based reasoning (Costa & Santos, 2019, p. 472) or sensemaking (Jones, Brookman, Williams, & Fraser 2020). Research in Switzerland (Resnikoff et al., 2015; Delémont, Bitzer, Jendly, & Ribaux, 2018) and the Netherlands (de Gruijter & de Poot, 2019) has identified the emergence of an *intelligence culture* informing crime scene examination (i.e., one that is focused on identifying patterns between crimes); however, the extent to which this has developed in other jurisdictions around the world requires further research. Changes are evident in some jurisdictions and in some types of crime scene examination. For example, in the United States, fire investigation has been part of a ‘paradigm shift in professional knowledge in the forensic sciences from experience-based to scientifically informed expertise’ (Dioso-Villa, 2013, p. 846) that has resulted in improvements in the reliability of fire origin and cause determinations (Lentini, 2015; Woodman, Ballantyne, Julian, & Spiranovic, 2021).

More empirical research is needed to unpack the complexity of CSE work and to identify how the critical decisions made by CSEs can be improved. A key tension between two epistemic cultures – a *scientific culture* (informed by science) and a *police culture* (centred on convictions) – is fundamental to this context and creates a central challenge. In some jurisdictions this is managed by a crime scene manager (CSM), who is typically a police officer, and whose role includes the central task of liaising between CSEs, police investigators and forensic specialists, as well as managers at the forensic science laboratory (Jones et al., 2020). While the responsibilities of this role are formally to manage the crime scene, it could be argued that informally a CSM’s most significant responsibility is to ‘translate’ information across the distinctive epistemic cultures of police investigators and forensic scientists.

Overall, it is apparent that crime scene work is complex. It requires sophisticated decision making, which is dependent upon knowledge of at least two complementary processes: criminal investigation processes (the domain of police) and the strengths and limitations of forensic disciplines (the domain of forensic science). This supports the view that CSEs are much more than technicians; they are the first decision makers in the forensic process. In fulfilling their overarching responsibility of *making sense* of the crime scene, they exercise agency and use their expertise (Wyatt, 2014). CSEs may need to have some contextual information about the case to make decisions effectively; however, this needs to be balanced with the need to avoid the negative effects of contextual and/or confirmation bias (Thompson, 2011; Gardner, Kelley, Murrie, & Dror, 2019; Jones et al., 2020). Furthermore, their decisions and interpretations lay the foundation for all subsequent interpretations by the

various actors in the criminal justice system. Therefore, they make a fundamental contribution to the jigsaw puzzle that comprises the forensic evidence informing the investigation, and to the construction of future narratives about the crime.

Forensic Laboratory

The next institutional location in the trajectory from crime scene to court is the forensic laboratory. The social processes in the laboratory have rarely been the subject of scrutiny due to the representation of scientific and technical processes as error free. Costa and Santos (2019) refer to a *bubble culture* that serves to maintain the boundaries between the laboratory, the police investigators and the courtroom. However, with the recognition that the scientific process is not technologically determined but rather is a social process that involves actors interpreting data and making decisions in various laboratory phases, a number of potential problems have been identified. Figure 5.2 depicts the phases of the forensic process at the crime scene (scene examination, collection and submission) and at the laboratory (receipt, triage, examination, analysis, interpretation and reporting).

INSERT FIGURE 5.2 HERE – ‘The forensic process: Phases at the crime scene and the laboratory’

In each phase, decisions are made that affect the trajectory of the trace through the laboratory and the construction of narratives about the forensic evidence. First, laboratory processes are influenced by the types and amount of physical evidence and traces received, as well as when and in what condition they are received. Research has shown that police investigators are often more focused on traces that can lead to the identification of a suspect than traces with other informational value (such as detecting a crime series or assisting in reconstructing the event), and this may influence the decisions they make about what items to submit to the laboratory (de Griujter & de Poot, 2019; Bitzer, Ribaux, Albertini, & Delémont, 2016). In some cases, traces are not sent to the laboratory because the police investigator has not identified a suspect, or they assume that the results will not be received in time to inform the investigation (Strom & Hickman, 2010). The receipt of large amounts of material also increases case workloads and subsequent backlogs at the laboratory, ultimately leading to delays in analysis and the reporting of results to police investigators (Strom & Hickman, 2010; Crouse et al., 2020). Reviews of forensic science in the United Kingdom and the United States have repeatedly identified backlogs and timeliness of results as significant problems for forensic services and law enforcement (NIJ, 2004; National Audit Office, 2003; American Society of Crime Laboratory Directors, 2004; Julian, 2005).

The existence of *evidence* backlogs (Robertson, 2012) points to the need for effective and efficient triage processes to avoid flooding the laboratory with unnecessary items and to improve timeliness in delivering results (Roux et al., 2014). An effective triage process focuses the forensic examination on relevant traces; that is, traces that are most likely to provide valuable information to police for investigative or intelligence purposes (Bitzer et al., 2016). Despite the critical role of triage at the laboratory, there is little systematic knowledge about the decisions made at this stage of the forensic process. Nevertheless, a number of

trends have been identified. Selection is often based on case urgency, the seriousness of the crime (e.g., violent crimes versus property crime), whether a suspect is known, or whether there has been a judicial direction to prioritise the case. Scientific aspects may also be considered; for example, the probability of extracting a usable profile from a degraded specimen (Raymond, Van Oorschot, Gunn, Walsh, & Roux, 2009). However, it appears that triage decisions are predominantly reactive rather than informed by a proactive strategy of triage management (Roux et al., 2014). Significantly, cultural beliefs and attitudes among forensic scientists also influence triage decisions. For example, research has shown that forensic scientists' decisions regarding rape kit testing are influenced by extra-legal factors such as stereotypes based on 'beliefs about sexual assault victims' character and credibility' (Campbell & Fehler-Cabral, 2020, p. 4).

With respect to the analysis phase, *context bias* has been identified as a problem in most forensic disciplines, including fingerprints (Dror, Charlton, & Péron, 2006; Dror, Péron, Hind, & Charlton, 2005; Stevenage & Bennett, 2017), bite mark analysis (Page, Taylor, & Blenkin, 2012; Osborne, Woods, Keiser, & Zajac, 2014), bloodstain pattern analysis (Taylor, Laber, Krish, Owens, & Osborne, 2016; Osborne, Taylor, Healey, & Zajac, 2016), handwriting analysis (Kukucka & Kassin, 2014), and even DNA which is held up as the 'gold standard' in scientific analysis (Dror & Hampikian, 2011). More recently it has been identified in disciplines such as forensic toxicology (Hamnet & Dror, 2020) and digital forensics (Sunde & Dror, 2019) that have been considered more dependent on technology and thus less likely to be influenced by human factors. The issue of context bias continues to be a matter of debate among forensic scientists. Some argue that forensic scientists should know as little as possible about the criminal case because this will influence the analysis and interpretation of the forensic data (Found, 2015; Dror, 2017; NRC, 2009). On the other hand, others argue strongly that contextual ignorance diminishes the capacity of the forensic scientist to interpret material and contribute meaningful, relevant and valuable results to the police for investigative or intelligence purposes (Champod, 2014). They argue that contextual information is necessary in order to ask the right questions and direct the workflow and resources at the laboratory to the analyses that will provide useful answers (Willis, 2009; see also Jones et al., 2020).

Finally, concerns have been expressed about the way in which forensic results are communicated in reports to the criminal justice practitioners who use this information to inform their decision making (Willis, 2009; Howes, 2019). When forensic science is poorly communicated or misunderstood, it can lead to flawed decision making among those who use the findings in their roles in the criminal justice process (Howes, 2019). Generally, reviews of forensic science communication have found that forensic reports tend to be written for a scientific audience rather than for police, lawyers and judges. In practice, the content, format and style of reports written by forensic scientists varies considerably between jurisdictions – nationally and internationally. In England and Wales, the development of streamlined reporting has been viewed as highly contentious. While it was conceived as a response to the need to address efficiency goals, and intended for preliminary reports for the prosecution,

some have expressed concern that it has reduced the quality of reports and that these have been used in court although they were not designed for that purpose (Howes, 2019).

In terms of the life trajectory of the forensic evidence, forensic laboratories and the experts located within them have been described as ‘obligatory passage points in the making of meaning and value’ (Costa & Santos, 2019, p. 472). However, where and how forensic laboratories ‘fit’ in the criminal justice system is not clear cut. There are contradictory processes at play relating to, on the one hand, the independence of the scientific processes and the need for analysts to be removed from potentially biasing contextual information, and, on the other, the need to acknowledge that a greater integration between the forensic laboratory and other agencies in the criminal justice would improve communication and enhance the flow of forensic information from the crime scene to the court (Jones et al., 2020). Willis highlights the problematic consequences of the organisational and cultural separation of the laboratory from the other domains in the criminal justice system:

Most thinking forensic scientists will agree that their work is determined by the samples taken from the scene but many laboratory systems are operated with little connection between the scene and the laboratory. Such a system is unlikely always to communicate the truth. A system where an isolated laboratory carries out sophisticated analyses in complete isolation from the context of the case is unlikely ever to be in a position to ask all relevant questions and, in my opinion, can on occasion be said to be acting unethically. (2009, pp. 531–532)

While it is possible to identify two extreme positions in relation to this debate, suggestions have been made, and variously implemented, that address both concerns. Thompson (2011) proposed two ‘hybrid’ approaches: a ‘case manager model’ and ‘sequential unmasking’. A case manager model involves a separation of functions among forensic scientists so that they perform *either* the role of analyst *or* that of case manager. The case manager is fully informed about the investigation, works collaboratively with the police investigators, participates in decisions about what specimens to collect at crime scenes and what analytic tests to conduct, and manages the workflow at the laboratory. The role of the analyst, on the other hand, is to perform analytic tests and case interpretation as directed by the case manager while remaining blind to context (see also Laurin, 2013, who proposes an ‘independent case coordinator’ role). Sequential unmasking involves controlling ‘the sequence in which various analyses are performed in order to minimise the potential for contextual bias’ (Thompson, 2011, p. 126).

The Police Investigation

The police play a central role in the construction of forensic evidence. As noted by Costa and Santos (2019, p. 472), ‘everything they do will reflect on the life trajectories of the forensic evidence’. Nevertheless, little attention has been paid to how police investigators *actually* perceive and use forensic evidence (Cole, 2010b; Ludwig et al., 2014; Wilson-Kovacs, 2018). Overall, the limited studies that have been conducted have regularly found a ‘low level of usage for largely corroborative purposes by police with limited understanding of the potential of such technologies’ (Fraser & Williams, 2009b, p. 606). In a US study, Strom and Hickman (2010) found that DNA evidence was significantly underused and that there was not only an

evidence backlog at the laboratory, but also a ‘less well-known backlog of evidence that is collected by the police but never submitted to the laboratory for analysis’ (Cole, 2010b, p. 376). Delays in receiving results – especially DNA results – have led to a tendency for police investigators to make use of forensic evidence only in the later stages of an investigation. For example, research has shown that in homicide cases, laboratory results are often not provided to investigators until the case has been solved (Schroeder & White, 2009).

Jurisdictional variations in policies and standard operating procedures, as well as cultural differences, affect decisions made by police with respect to the selection, submission and use of forensic evidence. Qualitative research that has attempted to understand police decision making with respect to forensic evidence has shown that various forms of bias can be identified in these decisions. For example, stereotypes about sexual assault victims influence police decisions to submit sexual assault kits to the forensic laboratory (Campbell & Fehler-Gebal, 2018, 2020). Legislative requirements and previous experiences also influence what types of forensic evidence are most likely to be used by police, as well as when and how such evidence will be considered valuable in the criminal investigation (Brookman et al., 2020; Brown, 2021). In Australian research, Brown (2021) found that experienced police investigators were able to make use of fingerprint and DNA evidence in volume crime investigations but were less familiar with other types of forensic evidence. In another study, Woodman, Julian, Spiranovic, and Ballantyne (2020) examined how police had used chemical trace evidence in a sample of past cases. The study highlighted that by making connections between forensic and non-forensic evidence, chemical trace evidence had the potential to impact on multiple stages of a case and could be used to guide decisions that influence how the case progressed through the criminal justice system. For example, one investigator stated that it ‘supported and corroborated a victim whose own evidence was not very credible’ (Woodman et al., 2020, p. 7). Nevertheless, Wyatt (2014, p. 444) has noted that ‘existing work does not acknowledge ... the heterogenous usage of forensic science in investigative practices’.

How police understand and use forensic evidence also depends on the phases in a criminal investigation, as well as the extent to which such evidence is being used as intelligence for crime disruption and prevention strategies. These processes and phases are depicted in Figure 5.3.

INSERT FIGURE 5. 3 HERE – ‘Police investigation: Processes and phases’

The goal of a criminal investigation is ultimately for the police to assemble the information they produce through their lines of enquiry ‘into a narrative format that seeks to tell a story of “who did what to whom and why?”’ (Innes & Clarke, 2009, p. 547; see also Brookman et al., 2020). This is a complex social process that encompasses two aspects of police work – investigative and detective – that are rarely acknowledged as different from each other:

Investigative work involves enquiring into the circumstances of an incident, establishing the contours of what has happened and most importantly whether this might be coherent with the definitions set out in criminal law about what constitutes a range of different crime types ... [while] the detective

processes of police crime work ... focus upon the identification of potential suspects and constructing a legal case that evidences their agency and responsibility. In effect, investigative processes are concerned primarily with establishing a definition of the situation, whilst the detective processes focus upon the assignment of blame to culpable individuals (Innes & Clarke, 2009, p. 547).

The distinction is important because, while forensic evidence can be critical to both aspects of a criminal investigation, the meaning and value of the forensic evidence may be different in these two phases. For example, in a sexual assault case, DNA evidence may have some influence on the course of the investigation in the first stage; however, once the circumstances of the case are known and if what is at issue then moves to consent rather than identification, such evidence may no longer be useful in the investigation (Ludwig, 2016, pp. 71–72). Research has also shown that the sensemaking processes in the two phases differ. There is a shift from an abductive approach (an exploratory, proactive approach that involves testing unlimited hypotheses) that is consistent with a ‘hunch culture’ to hypothetico-deductive reasoning (an informed, reliable, reactive approach that assesses the probability of alternative hypotheses) that is more consistent with a ‘scientific culture’ (Ribaux & Caneppele, 2018; Baechler et al., 2020; Brown, 2021). Importantly, as the criminal investigation progresses, the meaning and value of the forensic evidence changes. Additionally, whether the forensic evidence is being used for investigative, intelligence or court purposes is a significant factor (Ross, 2015), with some jurisdictions increasingly focusing on its intelligence value as police organisations embrace intelligence-led policing models (Ribaux, Roux, & Crispino, 2017).

More research on criminal investigation processes is needed to develop a comprehensive understanding of how police understand and use forensic evidence in criminal investigations that are increasingly led by forensics (Fraser, 2008). To date, research has shown that addressing the problem of a limited use of forensic evidence by police requires frequent and routine two-way communication between forensic personnel, police investigators and prosecutors (Strom, Hickman, Smiley-McDonald, Roper-Miller, & Stout, 2011), enhanced training on the strengths and limitations of a range of forensic evidence types for police investigators and detectives (Brown, 2021), and a better understanding of the roles, responsibilities and policies of each stakeholder in the criminal justice system as a whole (Strom et al., 2011; Kelty, Julian, & Ross, 2013; Jones et al., 2020).

Legal System and Court

The legal system and the courts comprise the final step in the forensic process in a criminal investigation. The presentation of forensic evidence in court is the outcome of a lengthy, complex process that involves a wide range of actors who make decisions within the context of various organisations, each with their own policies, protocols and everyday practices; that is to say, they each have their own epistemic cultures. In the context of a European inquisitorial system, Costa and Santos (2019, p. 472) note that ‘prosecutors tend to assume a relatively passive posture towards the construction of forensic evidence’ and describe them as having an *office culture*. The key task of the prosecutor is ‘the assemblage of the forensic evidence by translating it to legal language’ (Costa & Santos, 2019, p. 472) to create the

‘prosecution’s legal story’ (Kruse, 2012, p. 306). By contrast, in their ethnographic study of homicide cases in an adversarial system, Brookman et al. (2020, p. 4) highlight the active role of prosecutors, as they necessarily select ‘from a vast array of information, what to tell and how to tell it’ in their quest to construct a plausible narrative to be presented in court. They rely heavily on forensic science to strengthen the credibility of their narrative vis-à-vis alternative narratives. Drawing on Chapman’s (2015) analogy of playing forensic science monopoly, the epistemic culture in an adversarial system could be described as a *game culture*; that is, one in which the goal is to gain the legal advantage.

The different epistemic cultures of forensic science and the law have been acknowledged and discussed by socio-legal scholars for many years (Jasanoff, 1993). The co-existence of a scientific culture and legal culture has been described as a ‘marriage of opposites’ leading to tensions and problems in the presentation of forensic evidence in court (Wonder, 1989). Research has shown, for example, that lawyers struggle to understand and use DNA evidence effectively (Cashman & Henning, 2012); that the defence does not make effective use of forensic evidence, in part due to a perception of forensic scientists as being ‘owned’ by (or *for*) the prosecution (Cashman & Henning, 2012); jurors have difficulty understanding the evidence presented by forensic scientists (Martire, Kemp, Watkins, Sayle, & Newell, 2013; Martire, Kemp, Sayle, & Newell, 2014; Goodman-Delahunty & Hewson, 2010b); and jurors’ expectations that forensic evidence should be presented in court leads to time-wasting and delays in delivering justice (Dioso-Villa, 2015b). Research has addressed these tensions by developing recommendations to improve communication between forensic scientists and fact-finders such as judges and jurors. For example, a number of suggestions have been made to improve forensic scientists’ capacity to present scientific evidence so that it is more likely to be understood by non-scientists and enhances the likelihood that the weight of the forensic evidence will be assessed appropriately in the case (Howes, 2017).

Other mechanisms to address the epistemic frictions between science and law – and to enhance the flow of communication between forensic scientists, police and members of the legal profession – have been suggested. These include recommendations for ‘translation’ mechanisms, such as ‘hot-tubbing’ (whereby experts simultaneously confer with each other in court); and pre-trial case conferences and pre-trial meetings between forensic scientists and lawyers (Kelty et al., 2013) to improve what is essentially the co-presentation of expert evidence by forensic scientists and legal practitioners during examination and cross-examination in court (Howes, 2015a). The use of intermediaries or advisors has also been suggested. Edmond (2015, p. 98) has recommended ‘independent advisory panels able to synthesise relevant research in order to provide authoritative independent advice to lawyers and judges on matters of controversy’. In Europe, the role of forensic coordinator or forensic advisor has been introduced in some jurisdictions to provide a global view of the forensic evidence to non-specialists (Bitzer, 2019a, 2019b). More specifically, such advisors ‘act as an advising body to the magistrate to improve communication between the various parties involved in the investigation: magistrates, police, crime scene investigators, and forensic experts’ (Bitzer et al., 2018, p. 177).

The Forensic Process in the Criminal Justice System

A comprehensive understanding of the role of forensic science in the criminal justice system requires a systemic approach that examines the forensic process from the crime scene to the court. Through such an approach it is possible to identify where errors occur – and where potential errors could occur – so that they can be addressed (Doyle, 2010). While there are differences between inquisitorial and adversarial systems in how forensic science is enacted, errors and weaknesses occur in both systems (Vuille, 2013). A number of these errors arise from the fragmented nature of the criminal justice system, which consists of various agencies that lack system integration. Egger (1984, as cited in Margot, 2011) has pointed to the problem of ‘linkage blindness’. The term refers to a lack of understanding of the larger picture, which results from failure to see the links between various elements of a system (see also Kelty et al., 2013). Others have highlighted the importance of ‘reciprocal knowledge’ among practitioners in forensic and investigative roles in order to collaborate productively (Ludwig et al., 2012); the need for communication between forensic laboratories, the police and prosecutors (Strom et al., 2011; Jones et al., 2020); and the need for ‘forensic literacy’ among all actors in the criminal justice system, given that the forensic process is increasingly a key component of most investigations (Innes & Clarke, 2009; Krone, 2012; Brown, Ross, & Attewell, 2014). There are also arguments for greater forensic literacy among practitioners who are not necessarily working within criminal justice agencies, but may be critical actors in some types of criminal investigation; for example, health professionals in emergency services (de Oliveira Musse, Santos, da Silva Santos, dos Santos, & de Melo, 2020), child protection workers and teachers (Kelty, Julian, Bruenisholz, & Wilson-Wilde, 2018).

Issues for Consideration

Professional Independence and System Integration

In an Australian case in 2008, Farah Jama was wrongfully convicted of rape. The conviction was based solely on DNA evidence, which was later found to be contaminated. The subsequent inquiry into the case found that the Victorian criminal justice system had overwhelmingly let Farah Jama down (Vincent, 2010). The wrongful conviction highlights a key tension in the forensic process and its enactment in the criminal justice system; namely, the need to balance professional independence with collaboration and information sharing to avoid the problem of ‘justice silos’ (Rayment, 2010; Krone, 2012; Kelty et al., 2013). Justice Vincent’s report stated:

For a number of obvious and excellent reasons, it is vital for the integrity of the system of justice upon which our society depends, that the forensic scientists and those responsible for the other parts of an investigation operate within their designated areas and maintain an appropriate professional distance and independence of each other. [This does not mean,] however, that they isolate themselves. (Vincent, 2010, p. 53)

Professional integrity and independence are critical. However, the effective use of forensic science in criminal investigations and prosecutions is dependent upon system integration; that is, where stakeholders in the criminal justice system understand each other's roles and responsibilities, share information as appropriate, and collaborate effectively to pose the right questions so that justice is achieved.

How can these two seemingly contradictory goals be achieved? Key to managing this tension is a better understanding of the appropriate boundaries for forensic disclosure; that is, an understanding of what should be disclosed *to* forensic experts, by whom and when, as well as what should be disclosed *by* forensic experts, to whom and when (Almazrouei, Dror, & Morgan, 2019). Equally necessary is a better understanding of the strengths and weaknesses of different organisational relationships between forensic laboratories and other stakeholders in the criminal justice system, and the ways in which these impact on the enactment of forensic science. The growth of lab-on-a-chip technology and other developments in digital forensics point to a future in which forensic laboratories are brought closer to the crime scene and become more integrated with policing (Casey, Ribaux, & Roux, 2019). What are the benefits and risks of such developments? Will this lead police investigators to adopt more scientific processes, so that forensic science becomes an invaluable building block for greater professionalism and objectivity in criminal investigations (Wilson, 2009)? Or will forensic scientists become the handmaidens of police (Cole, 2013)? There is an urgent need for more sociological and criminological research to address these questions (Wilson-Kovacs, 2018).

Identifying Stakeholders in the Criminal Justice System

If forensic science is understood as enacted differently by various stakeholders in the criminal justice system, this raises broader questions: Who are the key stakeholders in the criminal justice system? How do they understand – and engage in dialogue about – the social and justice implications of forensic science (Julian & Jendly, 2018)? To date, the tendency has been to limit the analysis of the forensic process to police, forensic scientists and members of the legal profession. What about other contexts – such as courts-martial in the defence forces – in which forensic science is used to provide answers to legal questions (Wilson, Gahan, Lennard, & Robertson 2019, 2020)? Additionally, if forensic awareness among first responders is critical to the investigative process and adjudication outcomes, who are the first responders for crimes such as sexual assault? Are they child protection workers, school teachers, emergency department staff in hospitals, general practitioners, nurses and/or victim advocates? What do we know about their levels of forensic literacy and how these influence criminal justice outcomes? What about victims and offenders? While there has been some research on their awareness of forensic science (e.g., Chopin, Beauregard, & Bitzer, 2020), little research has explored their views of the ways in which forensic science is used in the criminal justice system. Further, we can ask what counts as forensic science? To what extent are Indigenous knowledges understood as relevant and valuable within the non-Indigenous criminal justice system? In the Australian context, for example, do Indigenous trackers have the status of CSEs, given their high level of expertise? In short, what are the objectives of the criminal justice system from the perspectives of these different stakeholders? These broad questions encourage a critical examination of the purposes of forensic technology in the

criminal justice system and an opportunity to explore the use of forensic science from a human rights perspective, an approach that has been somewhat lacking to date (Donnelly, 2012).

Conclusion

Forensic science has a great deal of potential to contribute to justice. However, doing so requires an understanding and management of human and/or social factors as fundamental to the forensic process. Accordingly, this chapter has examined forensic science as a social process that takes place within the context of the wider institutional framework of the criminal justice system. It has adopted a systemic perspective in order to gain an understanding of the whole process, including the interactions within it, so that the right questions can be posed and addressed. A systemic perspective also makes it possible to identify how and where errors can occur within different parts and at different stages of the system, rather than viewing them as arising from discrete and independent events and individual actions. To counter the predominant 'faith in science' perspective that informs much commentary on the role of forensic science in criminal justice, critical forensic studies research and analysis of actual forensic practice is needed. Such research is informed by a systemic perspective, which recognises the criminal justice system as being comprised of numerous institutions with their own epistemic cultures and influences.

Discussion Questions

1. How is the concept of epistemic culture useful for understanding the role of forensic science in the criminal justice system?
2. Why is it important to adopt a systemic framework when analysing miscarriages of justice?
3. In what ways would greater levels of forensic literacy among police investigators contribute to criminal justice processes and outcomes?
4. To what extent should victims, offenders and other members of the public be viewed as stakeholders in the forensic process?
5. What are the strengths and weaknesses of technological advancements enabling forensic analyses to be conducted at the crime scene rather than at a forensic laboratory?

Further Reading

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Figure 5.1 The forensic process in the criminal justice system

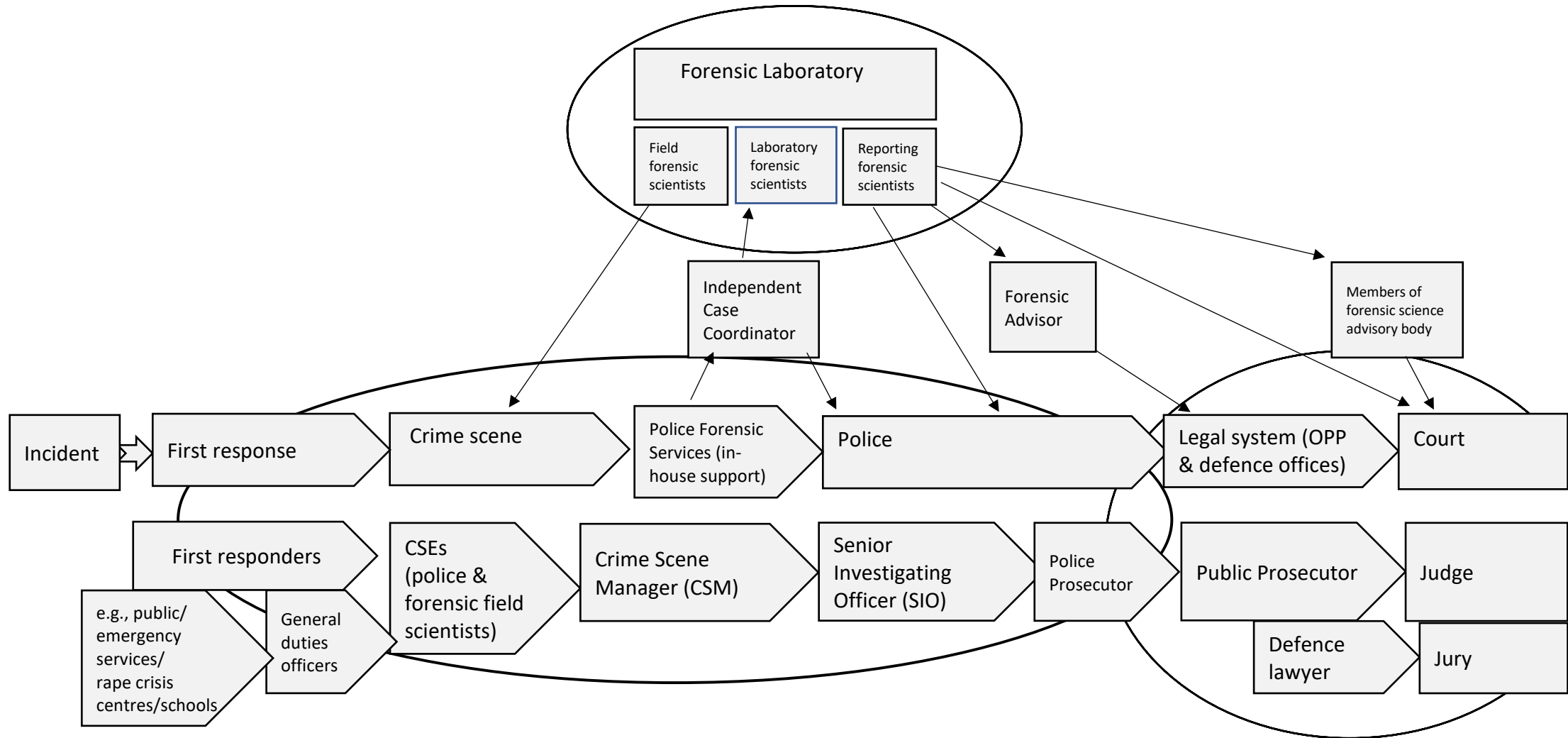


Figure 5.2 The forensic process – phases at the crime scene and the laboratory

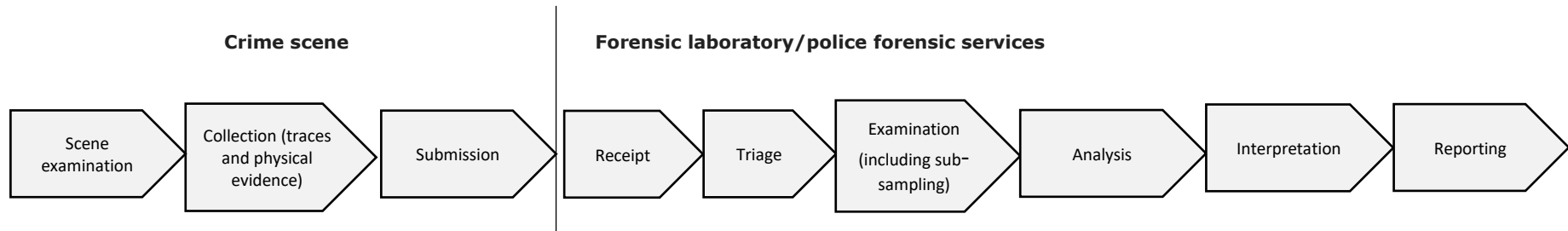
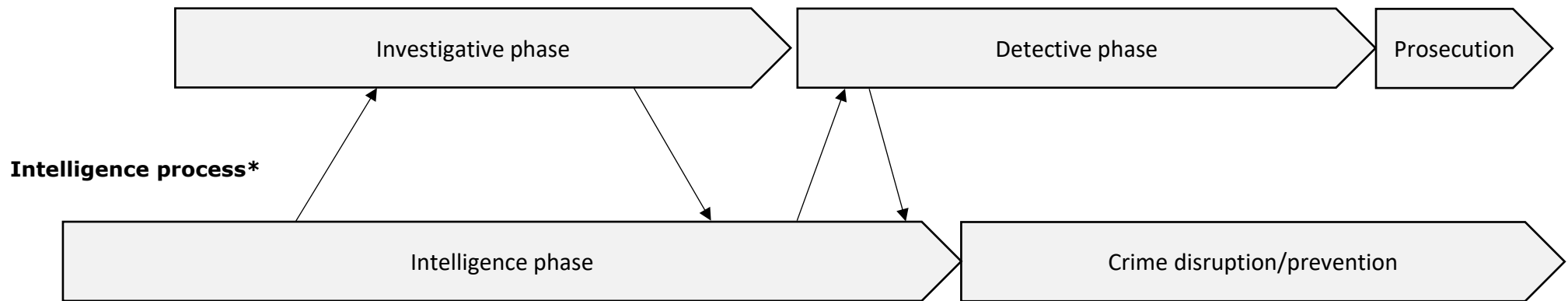


Figure 5.3 Police investigation: Processes and phases

Criminal investigation process



*Intelligence informs the investigation and therefore begins prior to the investigative phase.